

Indian Institute of Technology Gandhinagar

ES116 Principles and Applications of Electrical Engineering

Sem II, 2025-2026, Date: 31/01/2026

Assessment-I, Duration: 75 mins, Maximum Points: 40

Instructions:

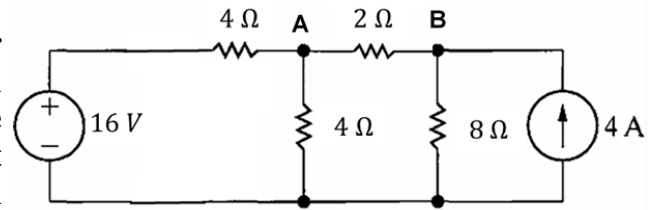
1. This assessment accounts for max. of 10 marks towards your total marks.
2. There are five questions in this assessment with their respective sub-parts.
3. It is recommended that you solve the questions in the given order.
4. Show all your work clearly. Partial credit for correct reasoning.

You are a systems engineer designing the power, sensing, and control units for a new lunar rover. You must apply circuit theory and signal concepts to ensure the rover survives the lunar night and operates its instruments correctly.

Q1. (3 points) The rover's primary solar array produces a current signal $i(t)$. The solar array current $i(t)$ fluctuates as the rover moves over rough terrain. The current signal is modeled as: $i(t) = 4 + 2\cos(2\pi t)$ A. This current flows into the main battery charging unit, which presents a constant resistive load of $R = 5 \Omega$.

1. (2 points) Calculate the RMS value of the current signal.
2. (1 points) Find average power delivered to the resistor.

Q2. (12 points) The Rover's central power system is shown in the given figure. Here, Radioisotope Thermoelectric Generator (RTG) is modeled as 16 V DC source with internal resistance of 4Ω connected to node A, while Solar array is modeled as a current source pushing 4 A into node B. The parallel resistance connected to node A and B of 4Ω



and 8Ω represent loads arising from heater and motor, respectively. The 2Ω resistor connecting Node A to Node B arises comes from a distribution cables.

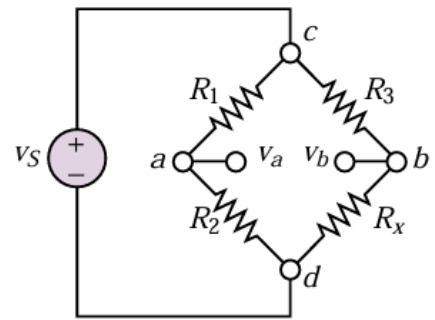
- A. (9 points) Calculate the power dissipated in the 2Ω resistor. Is the RTG delivering power or absorbing power in this specific configuration? Explain.
- B. (3 points) During a lunar dust storm, the output current from the Solar array (currently 4 A) could change significantly. Calculate the exact value of the solar current (I_{solar}) at which the current flowing through the Distribution Cable (Between A & B) becomes zero.

Q3. (9 points) The rover's long-range antenna is driven by a complex transmitter circuit with output terminals (A and B). To match the antenna to the transmitter, you perform two diagnostic tests on the output terminals (A and B):

- i) Open Circuit Test: You measure a voltage of 50 V across A-B,

- ii) Load Test: When you connect a $10\ \Omega$ test resistor across A-B, the voltage drops to 40V.
- (2 points) Calculate the Thevenin Equivalent Voltage and Thevenin Equivalent Resistance of the transmitter.
 - (1 point) The antenna impedance is variable. What resistance value (R_{load}) should the antenna be tuned to in order to receive Maximum Power? Justify your answer.
 - (2 points) The transmitter overheats if the total internal power dissipation exceeds 50 W. Is it safe to operate at the Maximum Power Transfer point? Prove your answer with a calculation.
 - (3 points) If not, calculate the specific antenna resistance R_{load} required to achieve the total power requirement of 50 W.
 - (1 points) Calculate the efficiency of the circuit for R_{load} value calculated in part D, where efficiency is defined as the ratio of power delivered to the load and total generated power (i.e P_{load}/P_{total}).

Q4. (12 points) A strain gauge (resistance varies with physical deformation) is mounted on the rover's suspension leg to measure the landing impact. It is placed in a Wheatstone Bridge configuration. Here, $R_1=R_2=R_3=100\ \Omega$, and R_x (gauge resistance at rest) = $100\ \Omega$. $V_S = 10\text{ V}$. The output of the bridge (Terminals a and b) is connected to Galvanometer which has an internal resistance of $R_{ab} = 50\ \Omega$ (not shown in the figure).



During a hard impact, the strain gauge resistance drops to $R_x = 80\ \Omega$. Calculate the exact current flowing through the Galvanometer and $V_{out} = V_A - V_B$ during the impact.

Q5. (4 points) You are installing a Gear Down indicator LED on the rover's dashboard. The circuit is powered by a 24 V voltage source. LED has a forward voltage drop of 4 V and current required is 0.1 A. You realized that the resistor is required to limit the current, so you installed one from your stock which are rated for 0.5 W. When you turned the switch on, the LED lit up for a split second, but the resistor instantly glowed red-hot and burned out with a puff of smoke. You only have a box of the same resistor values rated for 0.5 W.

Design a resistor network using only these components that provides the same equivalent resistance as needed by the LED and can safely handle the heat (i.e. Power Rating of resistor > Power Dissipation within the resistor). Draw the schematic of your solution (Series/Parallel arrangement) and prove it is safe by calculating the power dissipated by each individual resistor in your network.